TASK SPECIFICATION: P4-02 PHASE P4 — FINANCIAL LEDGER & TUITION OFFSET ENGINE

Wallet Service & Transaction Operations API

Git Branch: `feature/P4-02-wallet-api` · Target Pull Request: feature/P4-02-wallet-api

Core Objective & Technical Scope:
Enterprise financial engine: Real-time balance inquiry, paginated transaction statement, atomic peer-to-peer transfers, and idempotent payment handling.

Priority Level	Critical	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P4-02-wallet-api`
Upstream Dependencies	P4-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Implement WalletService supporting credit, debit, and atomic transfer methods.
2. Implement Idempotency-Key header filter guaranteeing safe retry semantics.
3. Create WalletController exposing /api/v1/wallet/ endpoints with OpenAPI documentation.
4. Execute multi-threaded stress tests to verify race condition resilience.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `wallet/application/*`
- `wallet/api/*`
- `tests`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Peer-to-peer transfers execute atomically inside a single database transaction.
- Zero double-spending under concurrent multi-threaded requests.

VERIFICATION & TEST COVERAGE STRATEGY

- WalletServiceTest
- ConcurrentTransferStressTest
- WalletControllerTest